
Duty of Life

Duty of Life is a full-stack sustainability platform designed to help people understand, track, and improve their environmental impact. The idea behind the project came from a simple observation: although many people care about sustainability, the tools available to support eco-friendly behavior are often scattered and disconnected.

Some tools calculate carbon emissions, others provide sustainability advice, while separate platforms manage CSR activities or environmental campaigns. Because these systems are fragmented, users rarely stay engaged for long. People may be aware that sustainability matters, but they often lack a single place where they can measure their impact, learn how to improve, participate in initiatives, and stay motivated.

Duty of Life solves this problem by bringing all these elements together into one unified platform. Instead of treating sustainability as a one-time calculation, the platform treats it as a continuous journey. Users can track their carbon footprint, receive practical eco-friendly suggestions, participate in CSR challenges, plant trees to offset emissions, share experiences with the community, and monitor their progress through scores and analytics.

The platform serves both individuals and organizations. Individuals can improve their daily habits and monitor their sustainability progress, while administrators can run structured CSR programs, verify participation, manage rewards, and encourage large-scale environmental action.

Core Value of the Platform

Duty of Life addresses three major challenges that commonly prevent people from adopting sustainable habits.

Awareness Problem

Many people do not realize how their everyday actions affect the environment. Activities such as commuting, electricity usage, and consumption patterns

contribute to carbon emissions, but these impacts are rarely visible. The platform helps users measure and visualize their environmental footprint.

Action Problem

Even when people understand that sustainability is important, they often do not know what steps to take. Duty of Life solves this by providing personalized advice, calculators, and practical recommendations.

Engagement Problem

Sustainability initiatives often fail because people lose motivation over time. The platform introduces gamification elements such as scores, badges, and leaderboards to keep users engaged.

Through these features, sustainability becomes:

- measurable through carbon calculations and scoring systems
 - actionable through advice, trackers, and challenge participation
 - motivating through recognition, leaderboards, and community interaction
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Technical Foundation

Duty of Life is built using a **MERN-style architecture**, which supports scalability and modular development.

Frontend Technologies

- React with Vite for fast UI development
- Material UI for responsive layouts and components
- Framer Motion for smooth UI animations

- Recharts for sustainability analytics visualization
- Axios for API communication

Backend Technologies

- Node.js with Express for REST API development
- MongoDB for storing application data
- Mongoose for schema management
- JWT for authentication
- bcrypt for password hashing
- express-async-errors for error handling

Database Models

The system uses structured models including:

- User – stores user profiles and sustainability scores
- Activity – records sustainability-related actions
- TreeOffset – tracks carbon offset through tree planting
- Challenge – stores CSR challenges created by admins
- Submission – stores user challenge submissions
- RewardDispatch – tracks rewards distribution
- Post – manages social posts
- Blog – stores long-form sustainability articles

Role-Based Access

The system supports two main roles:

User

- track emissions
- participate in CSR challenges
- write blogs and posts
- view leaderboard and analytics

Admin

- create CSR challenges
 - verify submissions
 - close contests
 - manage rewards
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Major Pages with Technical Structure

1. Home Page

The Home page acts as the public-facing entry point of the platform and introduces the vision behind Duty of Life.

Technical Implementation

- Implemented as a public **React route** with no authentication required
- Built using **Material UI grid system** for responsive layout

- **Framer Motion animations** used for hero banners and visual cards
- Navigation handled using **React Router**
- CTA buttons redirect users to authentication routes

Key Technical Points

- No backend API calls required
 - Pure UI rendering with reusable React components
 - Optimized for fast loading and user onboarding
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2. Authentication Page

The Authentication page manages secure login and registration.

Technical Implementation

- Single React component supports both login and registration modes
- Form state handled using React hooks
- Axios used for API communication
- JWT token stored in browser local storage

Backend Processing

- Password hashing using bcrypt
- JWT token generation for secure authentication
- Middleware verifies token for protected routes

3. Insights Page (Dashboard)

The Insights page acts as the **central sustainability dashboard** for the user.

Technical Implementation

- React `useEffect()` used for fetching analytics data
- Axios requests fetch score and emissions analytics

Frontend Components

- Score display card
- Badge display section
- Analytics charts using Recharts
- Lazy-loaded components for performance optimization

Backend Logic

- Aggregates user activity data
- Calculates sustainability score
- Generates emission trend analytics

4. Commute Page

The Commute page calculates **carbon emissions from travel**.

Technical Implementation

- Travel form collects distance and transport mode
- Form submission triggers emission calculation API

Backend Logic

- Applies emission factors based on transport type
- Calculates CO₂ emissions
- Suggests greener transport alternatives

Frontend Components

- travel input form
 - transport dropdown selector
 - emission result display
 - eco-friendly suggestion panel
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5. Advisor Page

The Advisor page provides **personalized sustainability recommendations**.

Technical Implementation

- User inputs lifestyle habits such as electricity usage and diet
- Data sent to backend advisory API

Backend Processing

- Rule-based recommendation system
- Calculates potential emission reductions
- Generates eco-friendly advice

Frontend Components

- habit input form
 - advice response section
 - reduction suggestions list
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6. Tree Offset Page

The Tree Offset page allows users to **offset carbon emissions through tree planting**.

Technical Implementation

- User enters emission amount and trees planted
- API calculates carbon offset value

Backend Logic

- Calculates CO₂ absorbed per tree
- Updates user sustainability score
- Stores tree planting records

Frontend Components

- offset calculation form
 - result summary display
 - offset progress dashboard
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7. CSR Challenge Page

The CSR page manages **environmental challenges and contests**.

Backend Logic

- Stores challenge metadata
- Validates submission deadlines
- Prevents duplicate submissions
- Tracks participant data

Frontend Components

- challenge creation form (admin)
 - challenge listing cards
 - proof upload interface
 - challenge metadata display
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8. Leaderboard Page

The Leaderboard page motivates users through **ranking and competition**.

Technical Implementation

Backend Logic

- Fetches all users
- Sorts by sustainability score
- Returns ranked list

Frontend Components

- podium-style top users display
 - searchable leaderboard list
 - ranking badges
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9. Blogs Page

The Blogs page supports **knowledge sharing and sustainability education**.

Technical Implementation

Backend Logic

- Stores blog content in MongoDB
- Supports tagging and metadata
- Returns sorted blog lists

Frontend Components

- blog writing panel
 - blog listing interface
 - detailed blog viewer
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10. Social Page

The Social page builds a **community around sustainability**.

Technical Implementation

Backend Logic

- stores user posts
- returns chronological feed
- manages post metadata

Frontend Components

- post composer
 - social feed display
 - quick carbon calculator
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11. Admin Page

The Admin page manages **CSR verification and reward operations**.

Technical Implementation

Backend Logic

- verifies user challenge submissions
- selects winners
- generates reward records
- prevents duplicate reward distribution

Frontend Components

- submission verification panel
- challenge closing interface
- reward dispatch tracking

Backend Functionality Explained Simply

The backend acts as the **core processing layer** of the platform.

Whenever a user performs an action, the frontend sends data to the backend. The backend then:

- validates the request
- applies business logic
- stores the data in MongoDB
- returns the result to the frontend

Examples include:

- calculating travel emissions
 - storing blog content
 - verifying CSR challenge submissions
 - computing sustainability scores
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Why This Project Stands Out

Duty of Life stands out because it integrates multiple sustainability components into one system.

The platform combines:

- personal sustainability tracking
- carbon emission measurement
- tree-based environmental restoration
- CSR program management
- social interaction and engagement
- sustainability education through blogs
- motivation through gamification

By connecting these elements, the platform transforms sustainability from a passive concept into an **interactive and measurable journey** where users can track their impact, improve their habits, and contribute to real environmental change.
